

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Soil Moisture Sensor Display</title>
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f0f8ff;
      text-align: center;
      padding: 20px;
    }
    h1 { color: #333; }
    #moisture-value { font-size: 2.5em; font-weight: bold; color: #007bff; margin: 20px; }
    button { margin: 10px; padding: 10px 20px; border: none; border-radius: 5px; font-size: 1.2em; }
    #chart-container { margin: 20px auto; width: 90%; max-width: 600px; }
  </style>
</head>
<body>
  <h1>Soil Moisture Sensor</h1>
  <p>Current Moisture Level:</p>
  <div id="moisture-value">--</div>
  <button id="connect-btn">Connect to Arduino</button>
  <button id="download-btn">Download CSV</button>
  <div id="chart-container">
    <canvas id="moisture-chart"></canvas>
  </div>
  <script src="script.js"></script>
</body>
</html>

```

Fig. 6: HTML code

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index.html  styles.css  script.js  +  43c63
1  let port, reader, keepReading = true;
2  const csvData = [{"Timestamp", "Moisture Level"}];
3  const ctx = document.getElementById("moisture-chart").getContext("2d");
4  const chart = new Chart(ctx, {
5    type: "line",
6    data: { labels: [], datasets: [{ label: "Moisture Level", data: [], borderColor: "#007bff" }] },
7    options: { scales: { x: { title: { display: true, text: "Time" } }, y: { beginAtZero: true } } }
8  });
9
10 document.getElementById("connect-btn").addEventListener("click", async () => {
11   try {
12     port = await navigator.serial.requestPort();
13     await port.open({ baudRate: 9600 });
14     const decoder = new TextDecoderStream();
15     const inputStream = port.readable.pipeThrough(decoder).getReader();
16     reader = inputStream;
17     readSensorData();
18   } catch (err) { console.error("Connection failed: ", err); }
19 });
20
21 document.getElementById("download-btn").addEventListener("click", () => {
22   const csvContent = "data:text/csv;charset=utf-8," + csvData.map(row => row.join(",")).join("\n");
23   const link = document.createElement("a");
24   link.href = encodeURI(csvContent);
25   link.download = "moisture_data.csv";
26   link.click();
27 });
28
29 async function readSensorData() {
30   while (keepReading) {
31     const { value, done } = await reader.read();
32     if (done) break;
33     const moistureValue = parseInt(value.trim(), 10);
34     if (!isNaN(moistureValue)) {
35       document.getElementById("moisture-value").textContent = moistureValue;
36       const time = new Date().toLocaleTimeString();
37       csvData.push([time, moistureValue]);
38       updateChart(moistureValue, time);
39     }

```

Fig. 7: JavaScript code for port connection and sensor data updating

Once you stop learning, you start dying.

—Albert Einstein

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